

QGreenFleet — Technical Fleet Optimization Report

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Fleet: 20 vessels, 5 routes

Engine: QIEA + QPSO (100 generations recorded)

1. Summary

KPI	BAU Baseline	Recommended plan	Delta (plan – BAU)
Fuel cost (\$)	\$3,078,567	\$1,223,204	–\$1,855,362 (reduction, –60.3%)
WtW GHG (t CO ₂ e)	16,852 t	3,371 t	–13,481 t (reduction, –80.0%)
OPEX (\$)	\$5,456,967	\$3,407,748	–\$2,049,219 (reduction, –37.6%)
Demand satisfied	100.0%	100.0%	
Total schedule delay	0.0 h	0.0 h	
Deployed vessels in CII A–C*	6 / 6	5 / 5	
All constraints met	yes	yes	

Feasible: no demand deficit, schedule delay, fuel-availability, CII or vessel-availability violations.

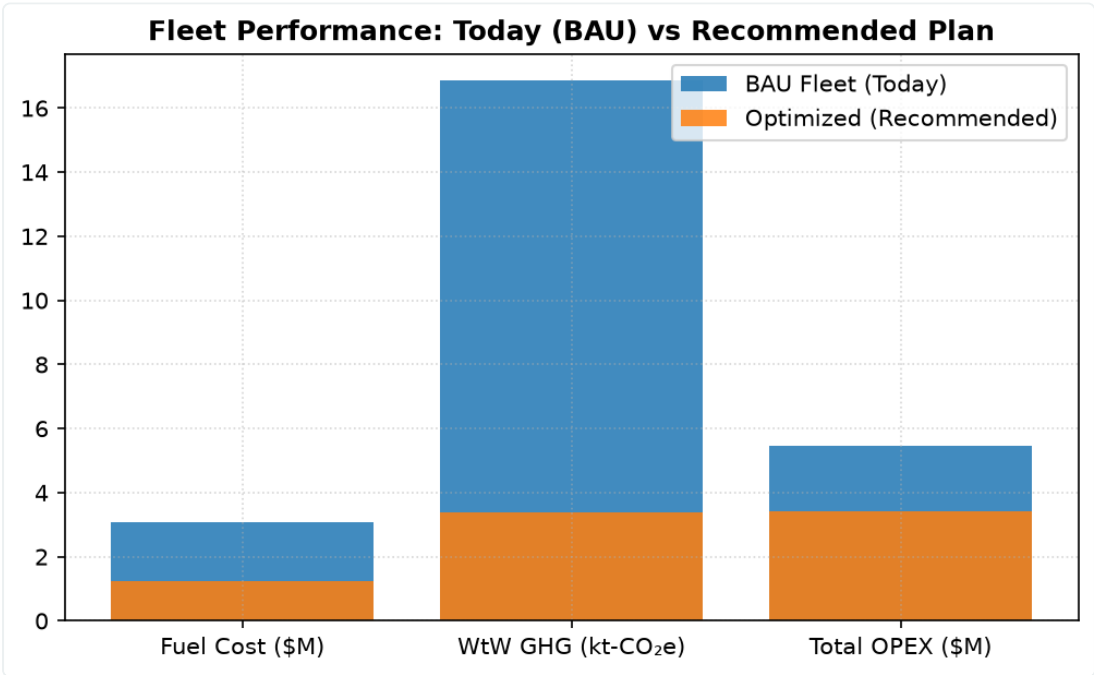


Figure 1: BAU baseline vs recommended plan.

2. Pareto Frontier & Trade-Off Surface

22 non-dominated solutions with objective values were supplied.

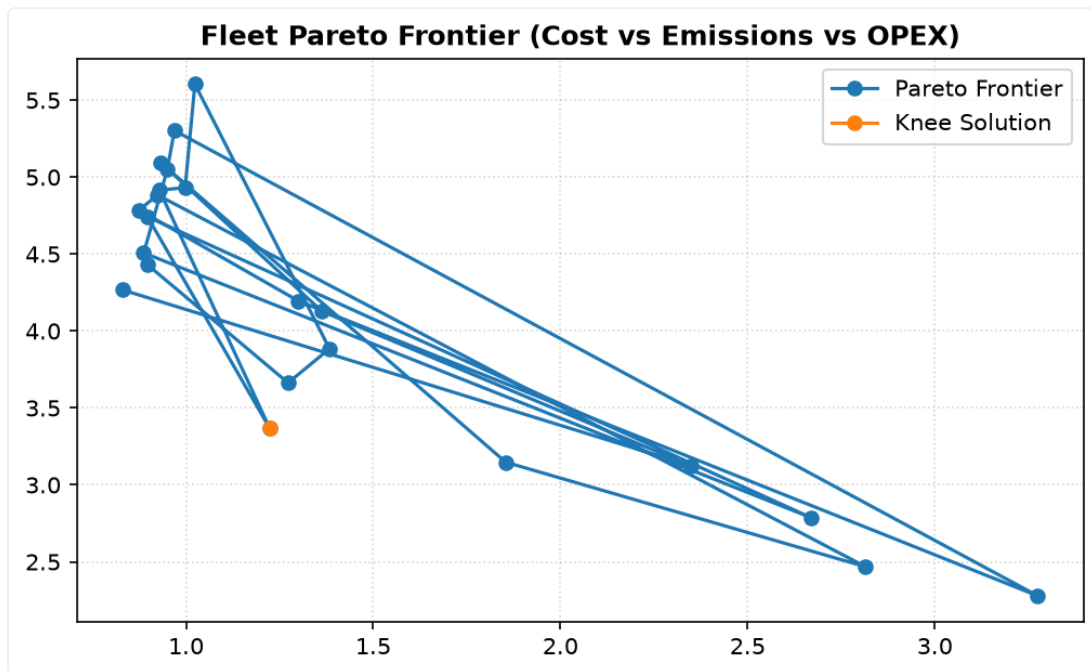


Figure 2: Pareto front: X = fuel cost (\$M), Y = WtW GHG (kt CO₂e), marker size = OPEX. Star marks the knee.

3. Recommended Deployment Plan

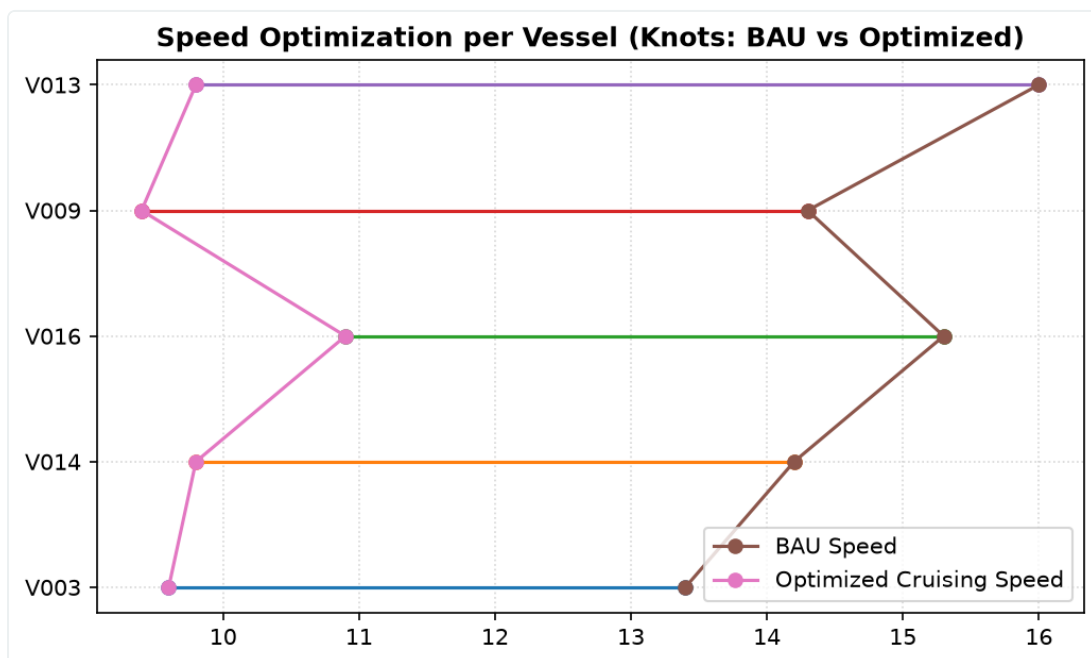


Figure 4: Per-vessel speed, BAU (grey) vs plan (green).

Vessel	Type	DWT	Route	Speed (kn)	Fuel	Fuel Cost	CO ₂ e (t)	CII*	Δ vs BAU
V000	tanker	234,684	Reserve	—	—	—	—	—	

Vessel	Type	DWT	Route	Speed (kn)	Fuel	Fuel Cost	CO ₂ e (t)	CII*	Δ vs BAU
									held in reserve (deployed in BAU)
V001	bulk	181,296	Reserve	—	—	—	—	—	held in reserve (deployed in BAU)
V002	tanker	156,015	Reserve	—	—	—	—	—	held in reserve (deployed in BAU)
V003	bulk	144,534	R0	9.6	HFO	\$195,336	1,069	A	slowed 3.8 kn
V004	container	58,986	Reserve	—	—	—	—	—	held in reserve (deployed in BAU)
V005	tanker	81,157	Reserve	—	—	—	—	—	held in reserve (deployed in BAU)
V006	tanker	91,266	Reserve	—	—	—	—	—	no change
V007	tanker	254,823	Reserve	—	—	—	—	—	no change
V008	container	127,355	Reserve	—	—	—	—	—	no change
V009	bulk	75,457	R3	9.4	HFO	\$203,026	1,111	A	deployed (reserve in BAU), slowed 4.9 kn
V010	container	41,988	Reserve	—	—	—	—	—	no change
V011	tanker	209,510	Reserve	—	—	—	—	—	no change
V012	bulk	105,310	Reserve	—	—	—	—	—	no change
V013	tanker	60,923	R4	9.8	MEOH_GREEN	\$605,816	97	A	deployed (reserve in BAU), switched to MEOH_GREEN, slowed 6.2 kn
V014	bulk	123,916	R2	9.8	LNG_DIESEL	\$119,496	548	A	deployed (reserve in BAU), switched to LNG_DIESEL, slowed 4.4 kn
V015	container	89,733	Reserve	—	—	—	—	—	no change

Vessel	Type	DWT	Route	Speed (kn)	Fuel	Fuel Cost	CO ₂ e (t)	CII*	Δ vs BAU
V016	bulk	30,393	R1	10.9	HFO	\$99,531	545	A	deployed (reserve in BAU), slowed 4.4 kn
V017	container	70,151	Reserve	—	—	—	—	—	no change
V018	tanker	45,742	Reserve	—	—	—	—	—	no change
V019	bulk	140,835	Reserve	—	—	—	—	—	no change

*CII band from attained/required CII using the IMO bulk-carrier rating boundaries (0.86/0.94/1.06/1.18) as a common approximation for all ship types.

4. Emission Profile & Decomposition

Lever (sequential attribution)	GHG change	
BAU total	16,852 t	
Deployment / assignment (plan vs BAU vessels, HFO, BAU speeds)	−8,071 t	reduction
Speed (plan vs BAU speeds, HFO)	−4,532 t	reduction
Fuel switching	−876 t	reduction
Shore power	−3 t	reduction
Plan total	3,371 t	−13,481 t overall

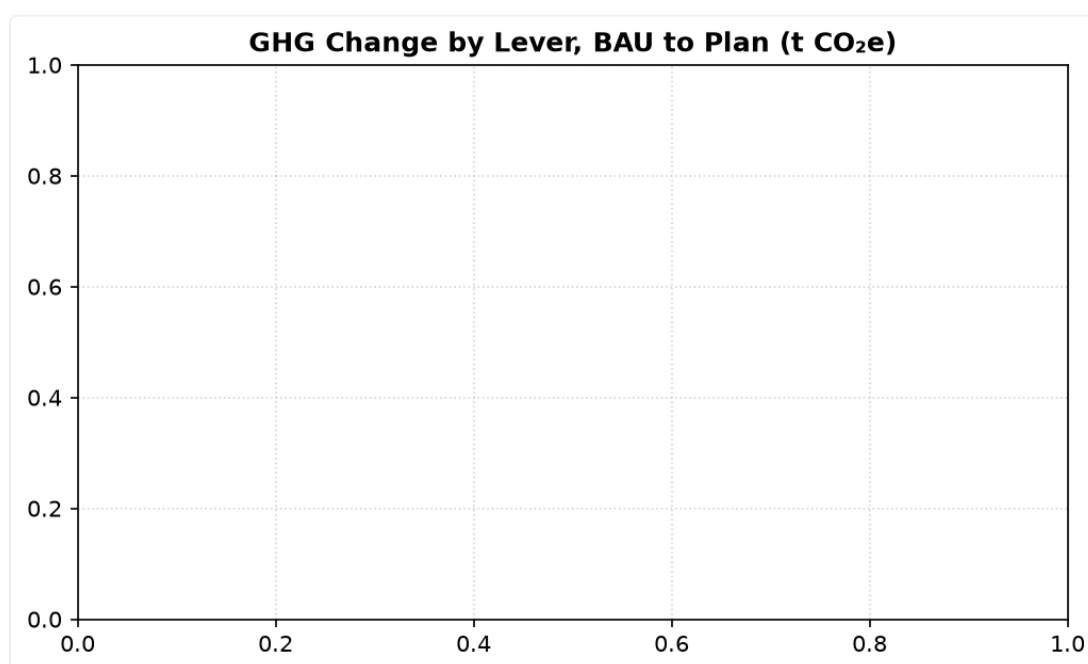


Figure 5: Exact additive attribution of the GHG change from BAU to the plan.

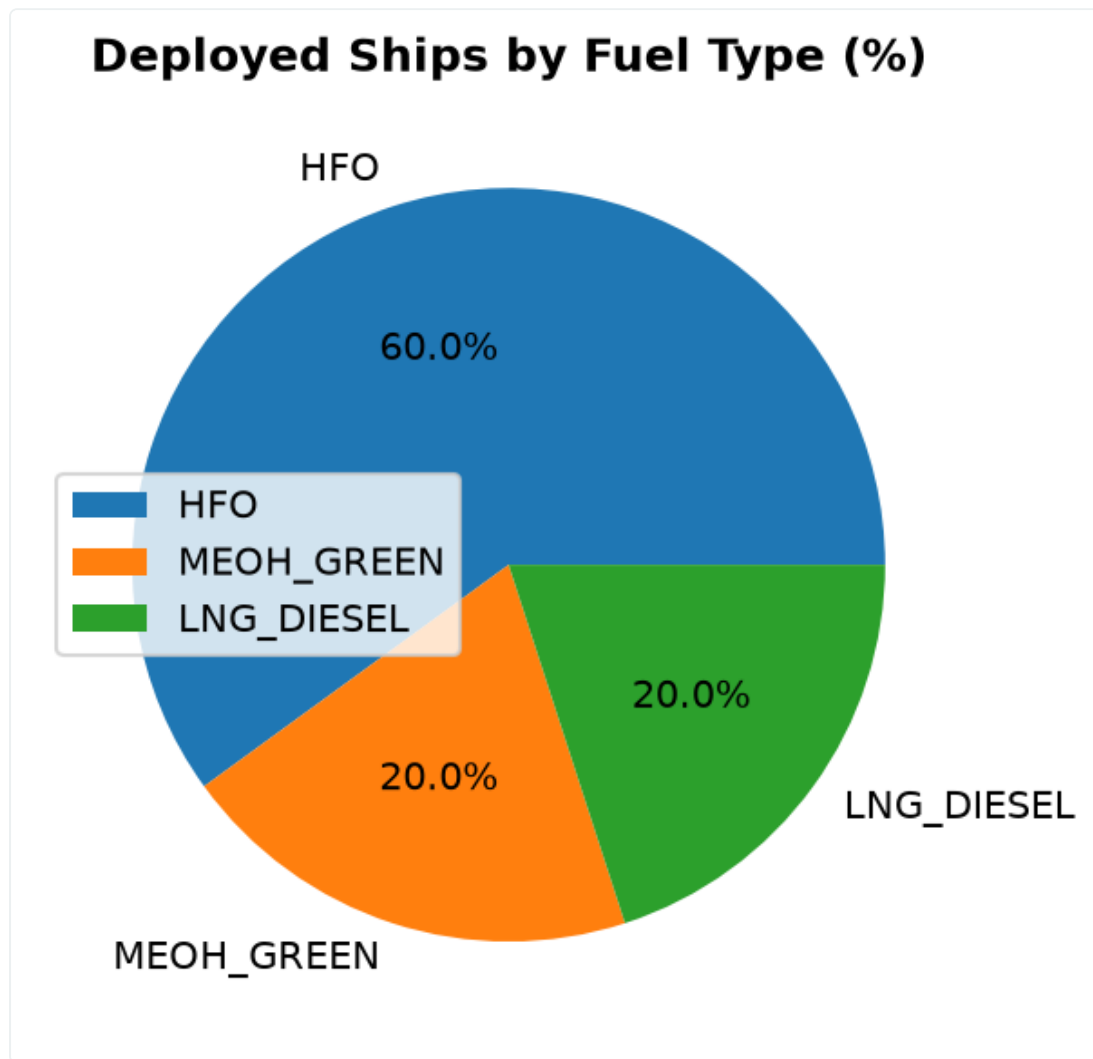


Figure 6: Deployed ships by fuel.

Emission factors: IMO 4th GHG Study (2020), FuelEU Maritime Regulation (EU) 2023/1805 Annex II, IPCC AR5 GWP₁₀₀.

5. Fuel Prediction Model

Stage 1 dataset: EU MRV THETIS (21,622 ship-years, ship-level 80/20 split).

Model	5-Fold CV RMSE	Test RMSE	Test MAPE	Test R ²	Status
Stage 1: MRV QPSO-XGBoost (held-out ships)	51.21 ± 4.32 kg/nm	45.23 kg/nm	26.3%	0.524	★ used
Stage 1: MRV QPSO-XGBoost (fleet-inference features)	—	52.93 kg/nm	34.5%	0.348	
MRV XGBoost, default hyper-parameters	—	45.35 kg/nm	26.5%	0.522	
Per-category median (no model)	—	57.39 kg/nm	36.9%	0.234	
		3.26 t/d	50.7%	-0.003	

Model	5-Fold CV RMSE	Test RMSE	Test MAPE	Test R ²	Status
Stage 2: voyage PhysicsBaseline	3.28 ± 0.08 t/d				not used — dataset has no learnable signal

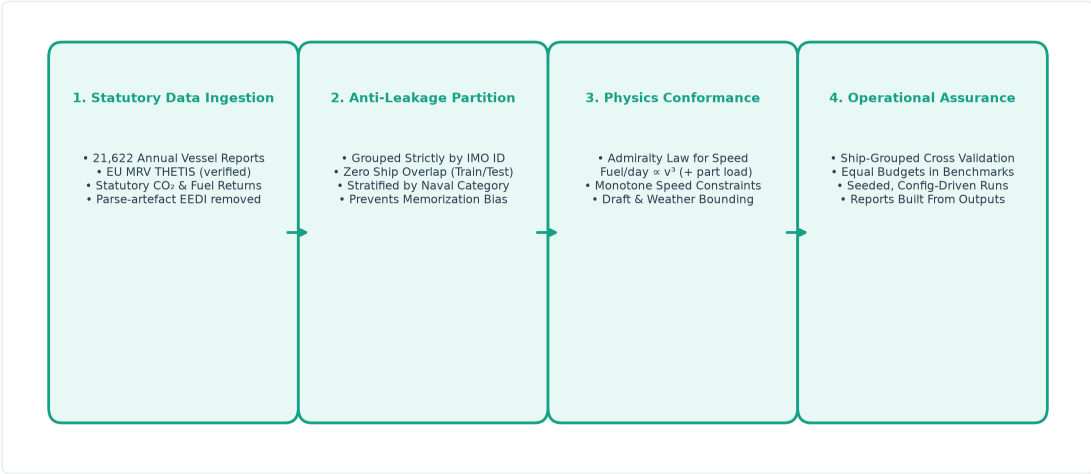


Figure 7a: Data provenance and validation.

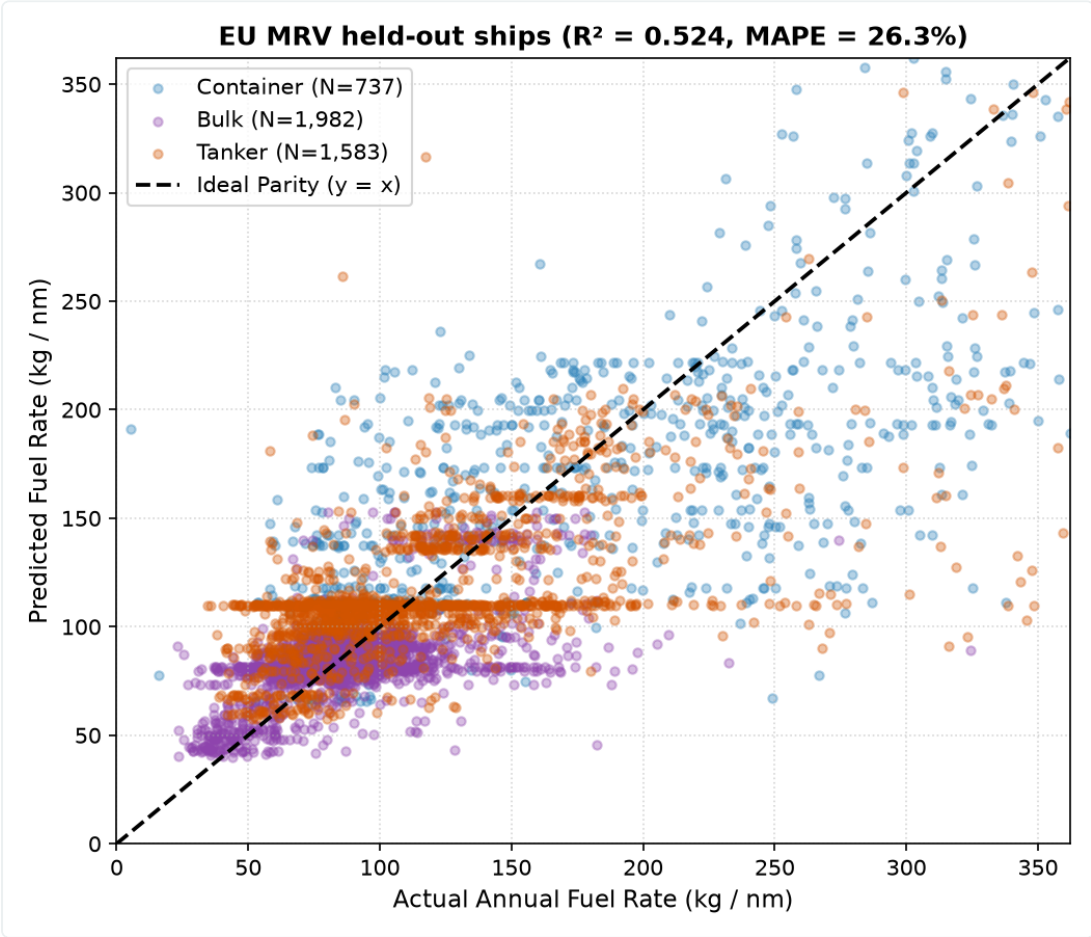


Figure 7b: Parity plot of the stage-1 MRV model on held-out ships.

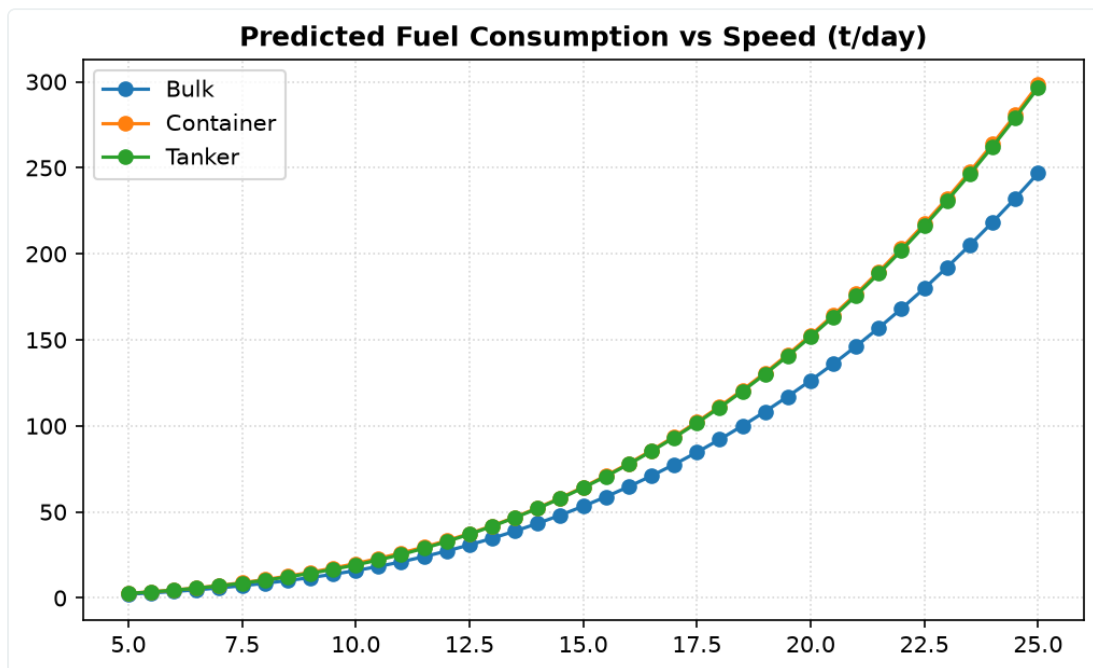


Figure 8: Predictor fuel consumption vs speed (draft 10 m, moderate weather).

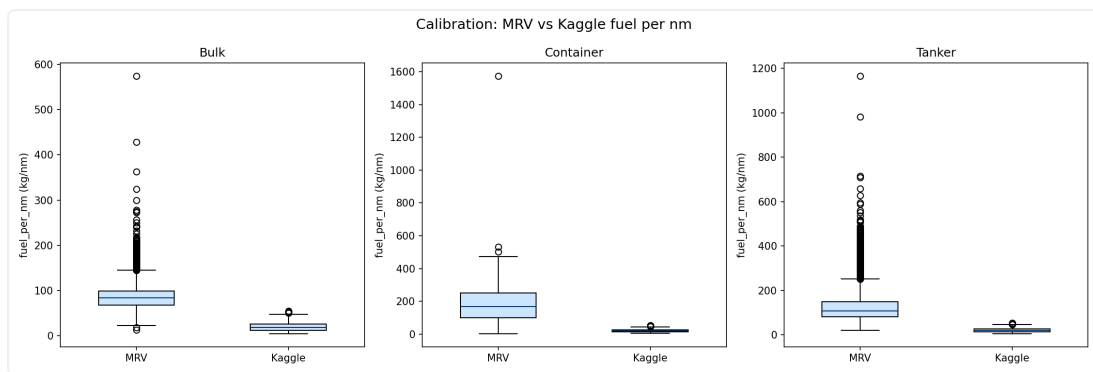


Figure 9: Calibration check against EU MRV fuel-per-nm distributions.

6. Optimizer Benchmark

Source: outputs/benchmark_results.csv; instances and budgets in configs/benchmark.yaml. Means over 5 seeds.

Metric	QIEA	GA	MOPSO	SA
Mean wall time (s) — S (5 vessels)	9.1	11.7	10.8	9.9
Mean wall time (s) — M (20 vessels)	19.5	24.8	19.0	12.0
Mean wall time (s) — L (50 vessels)	32.4	36.7	32.5	24.7
Mean wall time (s) — XL (100 vessels)	65.2	57.8	69.1	47.6
Mean hypervolume — S (5 vessels)	1.2865	1.2904	1.3302	0.1525
Mean hypervolume — M (20 vessels)	1.1327	0.9757	1.0973	0.3277
Mean hypervolume — L (50 vessels)	0.3803	1.1741	0.4056	0.0853
Mean hypervolume — XL (100 vessels)	0.1715	1.1826	0.1306	0.0359
Mean archive size — S (5 vessels)	30.6	61.4	95.0	2.4
Mean archive size — M (20 vessels)	12.8	84.4	9.8	3.2
Mean archive size — L (50 vessels)	3.6	6.8	4.2	2.0
Mean archive size — XL (100 vessels)	3.0	5.0	3.0	2.0
Mean feasible count — S (5 vessels)	22.0	43.8	20.4	0.8
Mean feasible count — M (20 vessels)	94.0	100.0	97.8	1.0
Mean feasible count — L (50 vessels)	66.0	44.2	69.0	1.0
Mean feasible count — XL (100 vessels)	74.8	100.0	83.8	1.0

Instance	QIEA wall time vs GA
S (5 vessels)	1.29× faster
M (20 vessels)	1.27× faster
L (50 vessels)	1.13× faster
XL (100 vessels)	1.13× slower

QIEA had the highest mean hypervolume on 1 of 4 instances.

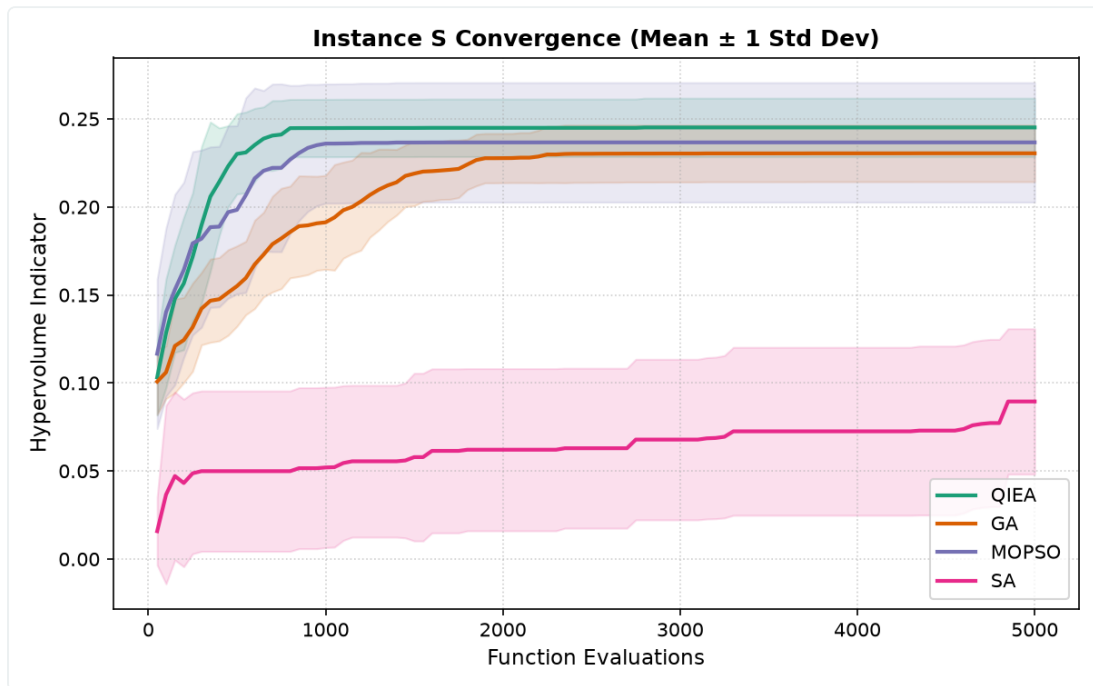


Figure 10: Hypervolume convergence across seeds.

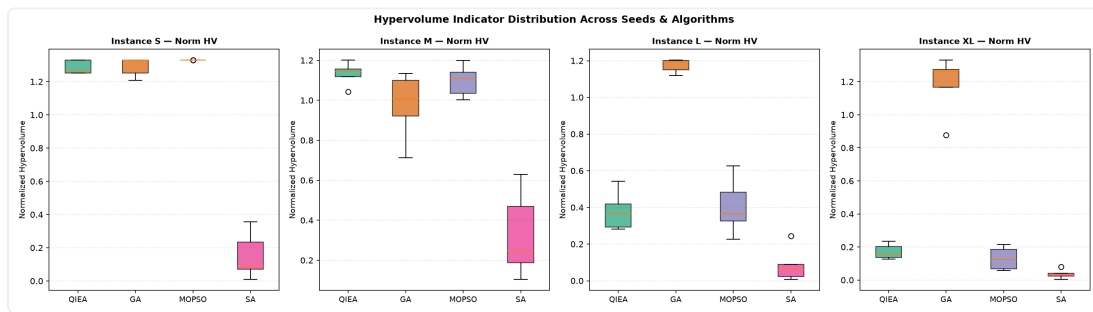


Figure 11a: Hypervolume distribution per algorithm.

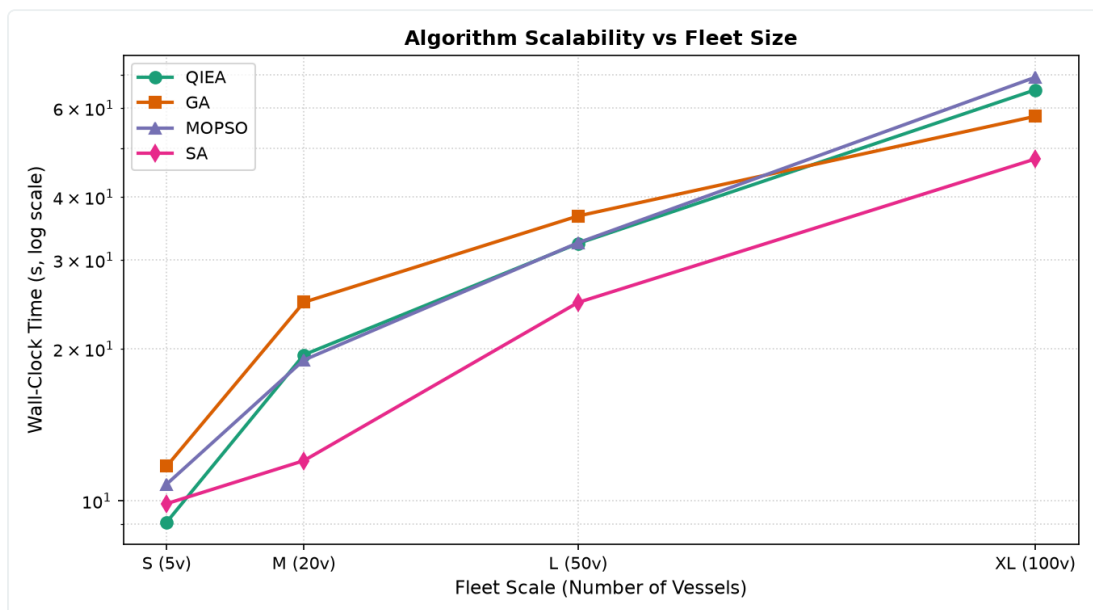


Figure 11b: Wall time vs fleet size.

7. Carbon Price Sensitivity

Swept carbon price \$0–\$200/t CO₂e. No crossover in the swept range: green methanol never reached the HFO share.

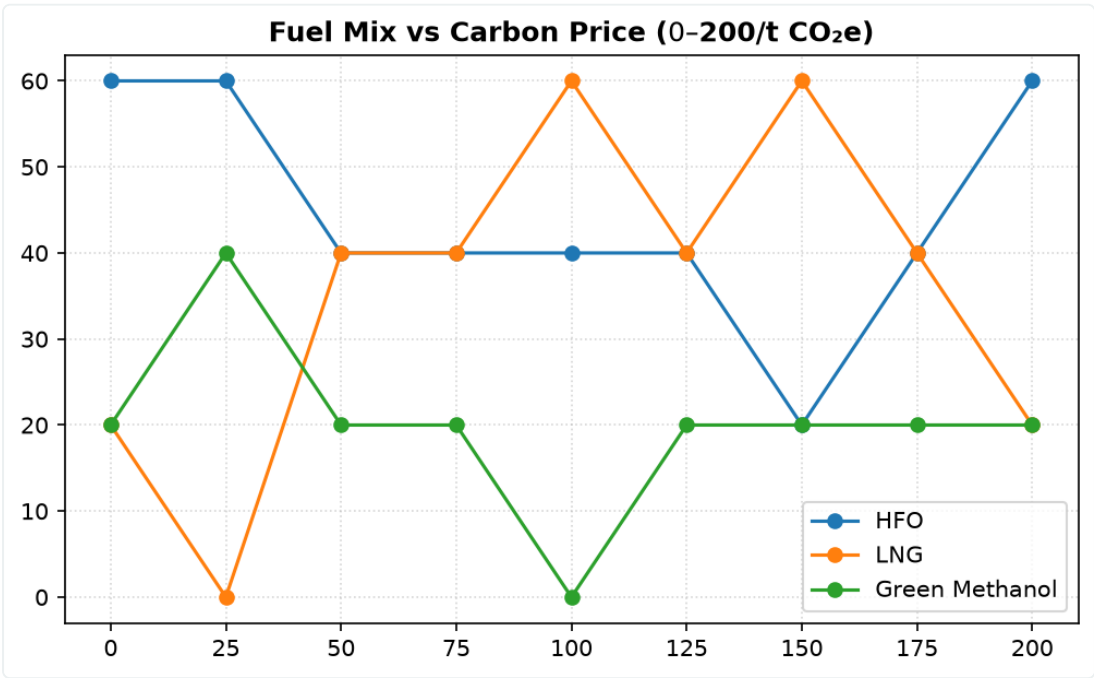


Figure 12: Share of deployed ships per fuel vs carbon price.

8. Algorithmic Methodology

QGreenFleet couples Quantum-Inspired Evolutionary Algorithms (QIEA, Han & Kim 2002) for combinatorial assignment, fuel selection, and shore power with Quantum-behaved Particle Swarm Optimization (QPSO, Sun et al. 2004) for continuous speed vectors. Non-dominated sorting and crowding distances maintain diversity in an external elitist archive. All algorithms execute as quantum-inspired surrogates on classical hardware.

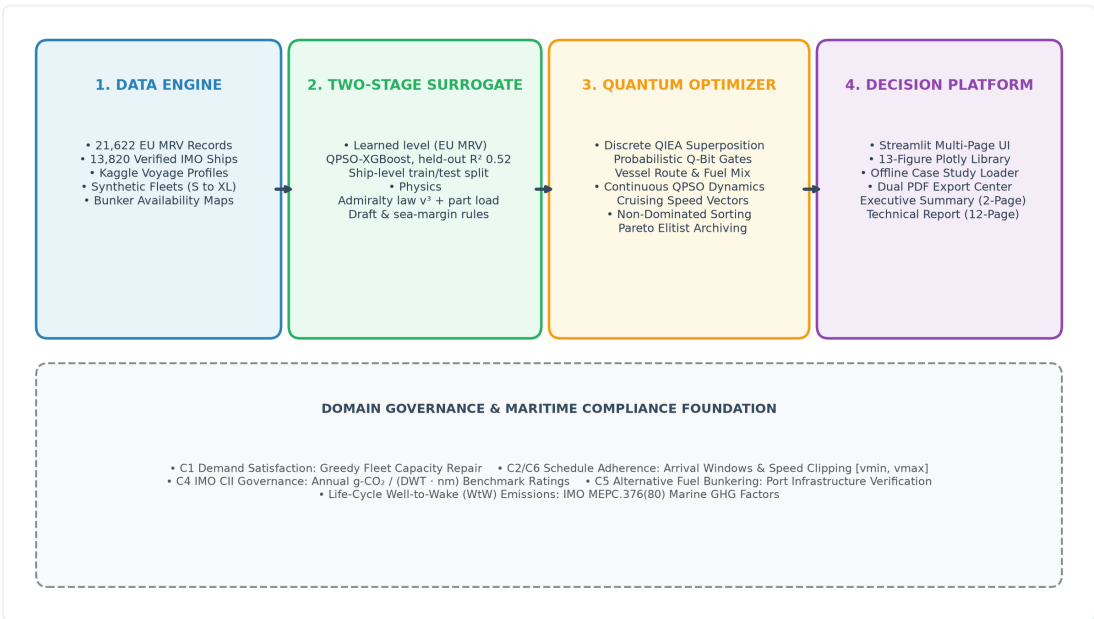


Figure 13: Hybrid QIEA+QPSO generational optimization loop.